

Customer Shopping Behavior Analysis

1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

2. Dataset Summary

- Rows: 3,900

- Columns: 18

- Key Features: -

Customer demographics (Age, Gender, Location, Subscription Status)

- Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)

- Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type) - Missing Data: 37 values in Review Rating column

3. EDA USING PYTHON

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using `pandas`.
- **Initial Exploration:** Used `df.info()` to check structure and `.describe()` for summary statistics.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes

Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
Yes	14	Venmo	Fortnightly
Yes	2	Cash	Fortnightly
Yes	23	Credit Card	Weekly
Yes	49	PayPal	Weekly
Yes	31	PayPal	Annually

- **Missing Data Handling:** Checked for null values and imputed missing values in the **Review Rating** column using the median rating of each product category.
- **Column Standardization:** Renamed columns to snake case for better readability and documentation.
- **Feature Engineering:**
 - Created **age_group** column by binning customer ages.
 - Created **purchase_frequency_days** column from purchase data.
- **Data Consistency Check:** Verified if **discount_applied** and **promo_code_used** were redundant; dropped **promo_code_used**.
- **Database Integration:** Connected Python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis.

4. EDA USING SQL (BUSINESS TRANSACTION)

We performed structured analysis in PostgreSQL to answer key business questions:

1. Revenue by Gender – Compared total revenue generated by male vs. female customers.

	gender	revenue
▶	Male	155438
	Female	75191

2. High-Spending Discount Users – Identified customers who used discounts but still spent above the average purchase amount.

	customer_id	purchase_amount
▶	2	64
	3	73
	4	90
	7	85
	9	97
	12	68
	13	72
	16	81
	20	90
	22	62
	24	88
	29	94
	32	79
	33	67
	35	91
	37	69
	40	60

3. Top 5 Products by Rating – Found products with the highest average review ratings.

	item_purchased	Average Product Rating
▶	Gloves	3.86
	Sandals	3.84
	Boots	3.82
	Hat	3.8
	Skirt	3.79

4. Shipping Type Comparison – Compared average purchase amounts between Standard and Express shipping.

	shipping_type	average_purchase
▶	Express	60.34
	Standard	58.41

5. **Subscribers vs. Non-Subscribers** – Compared average spend and total revenue across subscription status.

	subscription_status	total_customers	avg_spend	total_revenue
►	Yes	1033	59.42	61377
	No	2830	59.81	169252

6. **Discount-Dependent Products** – Identified 5 products with the highest percentage of discounted purchases.

	item_purchased	discount_rate
►	Hat	49.67
	Sneakers	49.66
	Coat	48.43
	Sweater	47.20
	Pants	46.43

7. **Customer Segmentation** – Classified customers into New, Returning, and Loyal segments based on purchase history.

	customer_segment	Number of Customers
►	Loyal	3085
	Returning	695
	New	83

8. **Top 3 Products per Category** – Listed the most purchased products within each category.

	item_rank	category	item_purchased	total_orders
►	1	Accessories	Jewelry	170
	2	Accessories	Sunglasses	159
	3	Accessories	Belt	159
	1	Clothing	Blouse	169
	2	Clothing	Shirt	168
	3	Clothing	Pants	168
	1	Footwear	Sandals	159
	2	Footwear	Shoes	147
	3	Footwear	Sneakers	145
	1	Outerwear	Jacket	163
	2	Outerwear	Coat	159

9. Repeat Buyers & Subscriptions – Checked whether customers with >5 purchases are more likely to subscribe.

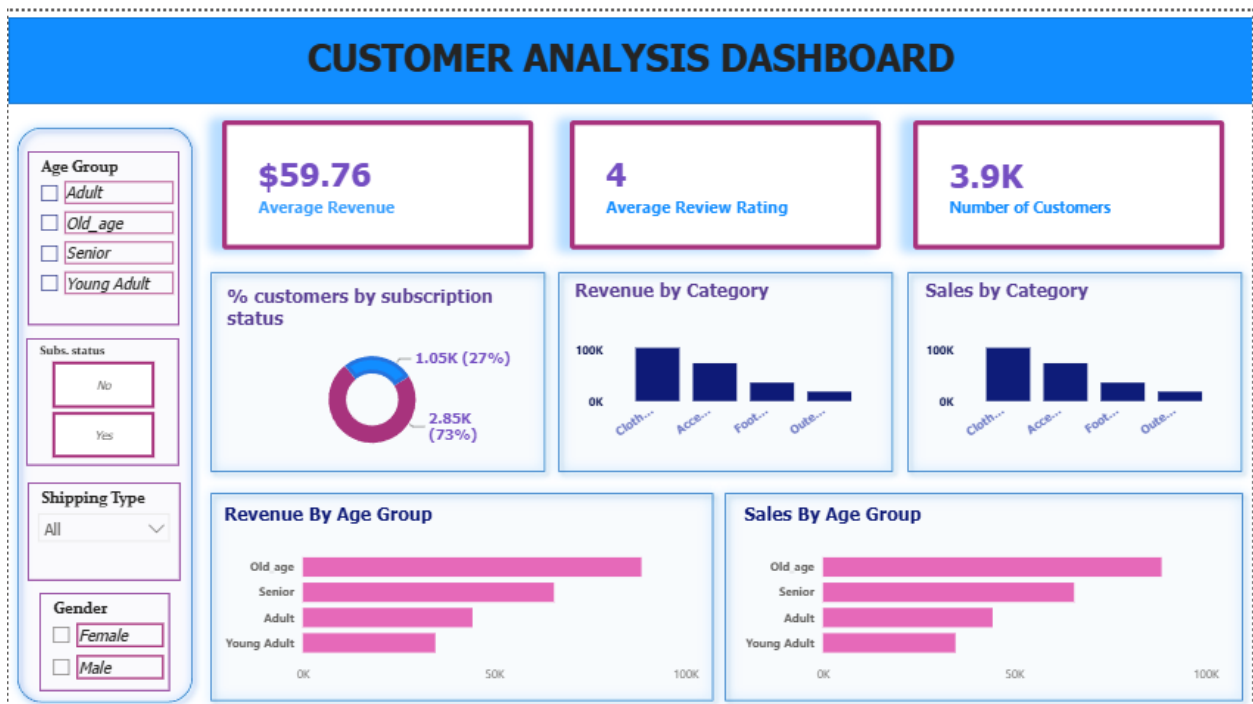
	subscription_status	repeat_buyers
►	Yes	939
	No	2503

10. Revenue by Age Group – Calculated total revenue contribution of each age group.

	total_revenue	Age_group
►	87638	OLD AGE
	64702	SENIOR
	44008	ADULT
	34281	YOUNG ADULT

5. DASHBOARD IN POWER BI

Finally, we built an interactive dashboard in **Power BI** to present insights visually.



6. BUSINESS RECOMMENDATIONS

- **Boost Subscriptions** – Promote exclusive benefits for subscribers
- **Customer Loyalty Programs** – Reward repeat buyers to move them into the “Loyal” segment.
- **Review Discount Policy** – Balance sales boosts with margin control.
- **Product Positioning** – Highlight top-rated and best-selling products in campaigns.
- **Targeted Marketing** – Focus efforts on high-revenue age groups and express-shipping users.